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zur Erlangung des akademischen Grades

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im Rahmen des Studiums

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in

Media Informatics and Visual Computing

by

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to the Faculty of Informatics

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Declaration of Authorship

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I hereby declare that I have written this thesis independently, that I have completely specified the utilized sources and resources and that I have definitely marked all parts of the work – including tables, maps and figures – which belong to other works or to the internet, literally or extracted, by referencing the source as borrowed.

I further declare that I have used generative AI tools only as an aid, and that my own intellectual and creative efforts predominate in this work. In the appendix “Overview of Generative AI Tools Used” I have listed all generative AI tools that were used in the creation of this work, and indicated where in the work they were used. If whole passages of text were used without substantial changes, I have indicated the input (prompts) I formulated and the IT application used with its product name and version number/date.

Vienna, January 1, 2001

Forename Surname

Danksagung

Ihr Text hier.

Acknowledgements

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Kurzfassung

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Abstract

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Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere.

1.1 Motivation

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turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus.

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1.2 Problem Statement

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1.2.1 Scope

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1.2.2 Out of Scope

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1.3 Contribution

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Related Work

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2.1 Typesetting Systems

Typst [Typ23] is a markup-based typesetting system, while LaTeX [Lam94] has been the established choice in academia for decades. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda

est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et.

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2.2 Positioning

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Using This Template

This chapter is the counterpart of the LaTeX class's own introductory chapter: it exercises every element a thesis is likely to need, so that anything looking wrong here can be spotted before you start writing. Delete it once you do.

3.1 Cross-references

Anything with a label can be referenced. Figure 3.1 is a figure, Table 3.2 a table, Equation (3.1) an equation, Algorithm 3.1 an algorithm and Listing 3.1 a listing. Sections work the same way: this is Section 3.1 inside Chapter 3. Citations go through the bibliography, as in [Lam94] and [Typ23].

3.2 Figures

3.2.1 A figure at full text width

Give the image `width: 100%` to fill the type block, and attach the label after the figure, as in Figure 3.1.



Figure 3.1: The TU Wien Informatics logo at the full width of the type block.

3.2.2 Several panels in one row

Papers often place two or three images side by side, each with its own caption and label. Wrap the panels in `subfigure-row` inside a normal figure: Figure 3.2a and Figure 3.2b are the panels of Figure 3.2.



(a) At the width of its column.



(b) At sixty per cent of it.

Figure 3.2: The same logo twice, to show how panels are lettered and referenced individually. This caption is long on purpose, and the list of figures shows the short form instead.

3.2.3 A diagram drawn in Typst

Diagrams need not be imported as images. The two below are drawn from the document source itself, so they pick up the body font and stay sharp at any zoom.

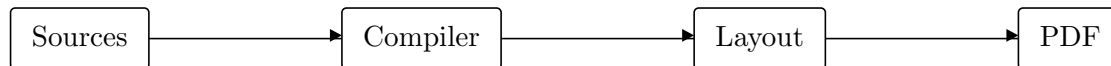


Figure 3.3: A pipeline drawn with boxes, rules and a polygon for the arrow head.

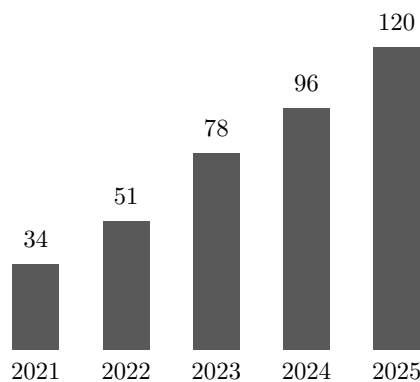


Figure 3.4: A bar chart computed from an array in the document source.

3.3 Tables

Rules come from `table.hline`; the booktabs look is a thick rule at the top and bottom and a thin one under the header. Table 3.1 is the plainest form.

Distribution	Unix	Windows	MacOS
TeX Live	yes	yes	(yes)
MacTeX	no	no	yes
MiKTeX	(yes)	yes	yes

Table 3.1: TeX distributions for different operating systems, with the recommended choice in **bold**.

Cells span rows and columns with `table.cell`, and a partial rule takes `start` and `end`. Table 3.2 is the class’s own example, carried over.

Position		
Group	Abbrev	Name
Goalkeeper	GK	Paul Robinson
Defenders	LB	Lucas Radebe
	DC	Michael Duberry
	DC	Dominic Matteo
	RB	Didier Domi
Midfielders	MC	David Batty
	MC	Eirik Bakke
	MC	Jody Morris
Forward	FW	Jamie McMaster
Strikers	ST	Alan Smith
	ST	Mark Viduka

Table 3.2: Adapted from the LaTeX guide at <https://en.wikibooks.org/wiki/LaTeX/Tables>. It shows cells spanning several rows and columns and a rule covering only part of the width.

3.4 Mathematical expressions

An expression can be set inline as $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, or displayed outside the text stream as

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

or as a numbered equation with

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad (3.1)$$

Equations align on `&`, and `cases` covers piecewise definitions.

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 & \nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{aligned} \quad (3.2)$$

$$\operatorname{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ +1 & \text{if } x > 0 \end{cases} \quad (3.3)$$

Matrices, determinants and large operators scale with their contents.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \det \mathbf{A} = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)} \quad (3.4)$$

The following exercises the symbol coverage of the maths font. Every row should render without a missing glyph.

Greek	$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\pi\rho\sigma\tau\upsilon\varphi\chi\psi\omega$
Greek, capital	$\Gamma\Delta\Theta\Lambda\Xi\Pi\Sigma\Upsilon\Phi\Psi\Omega$
Variants	$\epsilon\vartheta\phi\varpi\rho\varsigma$
Relations	$<\leq\ll\subset\subseteq\notin\equiv\approx\cong\neq\geq>\gg$
Operators	$\pm\mp\times\div\cdot\star\circ\oplus\otimes\wedge\vee\cap\cup$
Large operators	$\sum\prod\int\iint\oint\bigcup\bigcap\max_x$
Arrows	$\leftarrow\rightarrow\leftrightarrow\uparrow\downarrow\Leftarrow\Rightarrow\leftrightarrow\rightarrowtail\rightharpoonup$
Set theory	$\emptyset\varnothing\forall\exists\forall\exists\infty\partial\setminus\mathbb{C}$
Logic	$\neg\wedge\vee\Rightarrow\Leftrightarrow\models\perp\top$
Calculus	$\lim_{x\rightarrow 0}\frac{\sin x}{x}=1\quad\int_a^b f(x)\,dx\quad\frac{\partial^2 u}{\partial x\partial y}$
Delimiters	$\left(\frac{a}{b}\right)\left[\frac{a}{b}\right]\left\{\frac{a}{b}\right\}\left \frac{a}{b}\right \left\ \frac{a}{b}\right\ \lfloor x\rfloor\lceil x\rceil$
Accents	$\hat{a}\tilde{a}\bar{a}\acute{a}\grave{a}\check{a}\overline{AB}$
Blackboard, script	$\mathbb{N}\mathbb{Z}\mathbb{Q}\mathbb{R}\mathbb{C}\quad\mathcal{ABF}\quad\mathfrak{gh}$
Spacing and text	$a\,b\,c\,d\quad e\quad f\quad\text{plain text}\quad\limsup$

Table 3.3: A survey of the mathematical symbols the template renders.

3.5 Code listings

Fenced blocks are highlighted by language. Wrap one in `listing` to give it a number, a caption and an entry in the list of listings, as in Listing 3.1.

```
def fibonacci(n: int) -> int:
    """Return the n-th Fibonacci number."""
    a, b = 0, 1
    for _ in range(n):
        a, b = b, a + b
    return a
```

Listing 3.1: An iterative Fibonacci function in Python.

Short fragments stay inline as raw text: `git rebase --interactive`, or `typst compile --font-path fonts template/thesis.typ`.

3.6 Algorithms

Pseudo code is written as a list of steps, each with an indentation level, which keeps the line numbering continuous across nesting the way `algorithm2e` does.

Input: A scalar ε , a matrix $A = (a_{ij})$, a vector $\vec{b}$, and an initial vector $\vec{x}^{(0)}$

Output: $\vec{x}^{(n)}$ with $A\vec{x}^{(n)} \approx \vec{b}$

```

1  for  $k \leftarrow 1$  to maximum iterations do
2    for  $i \leftarrow 1$  to  $n$  do
3       $x_i^{(k)} = \frac{1}{a_{ii}} \left( b_i - \sum_{j < i} a_{ij} x_j^{(k)} - \sum_{j > i} a_{ij} x_j^{(k-1)} \right)$ 
4    end
5    if  $|\vec{x}^{(k)} - \vec{x}^{(k-1)}| < \varepsilon$  then break for
6  end
7  return  $\vec{x}^{(k)}$ 
```

Algorithm 3.1: Gauss–Seidel

3.7 Acronyms, a glossary and an index

Terms are declared once in `thesis.typ` and used with `gls`. The first use of an acronym spells it out and the rest give the abbreviation, so Portable Document Format (PDF) reads in full here and as PDF from now on. `gls` takes `plural` and `capitalize` for the other forms: editors, What You See Is What You Get (WYSIWYG). `acrfull` gives both forms wherever you need them: Comprehensive TeX Archive Network (CTAN).

A word is added to the index by marking a place in the text with `index-entry`. The same term can be marked as often as needed the entry collects the pages.

The three lists at the back are printed by `acronyms()`, `glossary()` and `index()`, and each lists only the terms actually used, as `\printglossaries` does.

3.8 Lists and quotations

- An unordered list,
- with a second item,
 - and a nested one.

1. An ordered list,
2. whose numbers follow automatically.

Term A description list pairs a term with its explanation.

Second term Which is what a glossary entry looks like.

3. USING THIS TEMPLATE

Programs are meant to be read by humans and only incidentally for computers to execute.

— Donald E. Knuth

Footnotes hang off the page they appear on.¹

¹Like this one.

Overview of Generative AI Tools Used

List here every generative AI tool used in the making of this thesis, where it was used, and — for passages taken over without substantial changes — the prompts and the product name and version of the application.

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Acronyms

CTAN Comprehensive TeX Archive Network. 13
PDF Portable Document Format. 13
WYSIWYG What You See Is What You Get. 13

Glossary

editor A text editor is a type of program used for editing plain text files. 13

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Bibliography

- [Typ23] Typst GmbH, “Typst: Compose papers faster.” [Online]. Available: <https://typst.app/>
- [Lam94] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.